

Homework 2

Darin Peries

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Problem 1

Part 1

To prove that any $A \in \mathbb{C}^{n \times n}$ follows the relation below,

$$n^{-1}\|A\|_2 \leq n^{-1/2}\|A\|_\infty \leq \|A\|_2 \leq n^{1/2}\|A\|_1 \leq n\|A\|_2$$

In order to prove that the relation holds for all $A \in \mathbb{C}^{n \times n}$, we need to rewrite the norm as $\|A\| = \sup\{\|Au\|, \|u\| = 1\} = \frac{\|Au\|}{\|u\|}$. The next step will involve the Holder inequality. This involves splitting the norm into two norms being multiplied, resulting in, for example the 1-norm,

$$\|Au\|_1 \leq \left(\sum_{i=1}^n |Au_i|^2\right)^{1/2} \times \left(\sum_{i=1}^n |1|^2\right)^{1/2} = \sqrt{n}\|Au\|_2$$

From the above expression, we can say that $\sup\{\|Au\|_1\} \leq \sqrt{n}\sup\{\|Au\|_2\}$ or $\|A\|_1 \leq \sqrt{n}\|A\|_2$. This gives us a part of the relation. The idea can then be extended to both the 2-norm and infinity norm as follows. $\|Au\|_2^2 = \left(\sqrt{\sum_{i=1}^n |Au_i|^2}\right)^2 \times 1 \leq \|Au\|_\infty \|1\|_1 = n\|Au\|_\infty$. Which can then be rewritten as $\|A\|_2 \leq \sqrt{n}\|A\|_\infty$ by taking the square root of both sides. For the 2-norm and 1-norm, we know that the $\|A\|_\infty \leq n\|A\|_1$ and an upper bound of the 2-norm is $\sqrt{\|A\|_1\|A\|_\infty}$ via the properties of the spectral radius covered in class, so we can then say $\|A\|_2 \leq \sqrt{\|A\|_1\|A\|_\infty} \leq \sqrt{\|A\|_1 n\|A\|_1} = \sqrt{n}\|A\|_1$. Lastly, $\|A\|_\infty \leq n\|A\|_1 \leq n\sqrt{n}\|A\|_2$. This gives us the following terms:

1. $\|A\|_1 \leq \sqrt{n}\|A\|_2$
2. $\|A\|_2 \leq \sqrt{n}\|A\|_\infty$
3. $\|A\|_2 \leq \sqrt{n}\|A\|_1$
4. $\|A\|_\infty \leq n^{3/2}\|A\|_2$

By scaling the bounds appropriately, say dividing by n , the relation can be upheld.

$$n^{-1}(\|A\|_2 \leq n^{1/2}\|A\|_\infty) \leq n^{-1/2}(n^{3/2}\|A\|_2 \leq \sqrt{n}\|A\|_1) \leq n\|A\|_2$$

Part 2

To prove that $\|x\| = \max_{i=1,\dots,n} w_i|x_i|$ is a norm on $\mathbb{R}^n$, we need to show that it holds the following:

1. $\|x + y\| \leq \|x\| + \|y\|$
2. $\|\lambda x\| = |\lambda|\|x\|$
3. $\|x\| = 0 \iff x = 0$

Triangle inequality

$\|x + y\| = \max_{i=1,\dots,n} w_i|x_i + y_i| = \max_{i=1,\dots,n} |w_i x_i + w_i y_i| \leq \max_{i=1,\dots,n} |w_i x_i| + \max_{i=1,\dots,n} |w_i y_i| = \|x\| + \|y\|$, via the subadditivity of the max function.

Positivity

A norm on $\mathbb{R}^n$ needs to be ≥ 0 . In order to prove this, the first step involves assuming $x = 0$. By assuming $x = 0$ where $x \in \mathbb{R}^n$, the resulting norm is as follows:

$$\max_{i=1,\dots,n} (w_i * 0, w_{i+1} * 0, \dots, w_n * 0) = 0$$

This shows that $\|x\| = 0 \implies x = 0$. Now the second step assumes that $\|x\| = 0$ and we need to prove that $x = 0$. Looking at the norm, $\max_{i=1,\dots,n} w_i|x_i|$ can only be negative if w_i is negative. Since w_i is a positive number, the norm being negative is not possible. Further more, the max function guarantees that the norm is non zero as long as some $x_i > 0$ for all i . Therefore, in order for $\|x\| = 0$, $x = 0$.

Homogeniety

Given some scalar $\lambda \in \mathbb{R}$,

$$\begin{aligned}\|\lambda x\| &= \max_{i=1,\dots,n} w_i |\lambda x_i| = \max_{i=1,\dots,n} |\lambda| w_i |x_i| \\ &= |\lambda| \max_{i=1,\dots,n} w_i |x_i| = |\lambda| \|x\|.\end{aligned}$$

Because the weights are all positive integers, and the max is taken over all the absolute values of x , the norm is greter than or equal to zero. Since all three postulates have been proven, $\|x\| = \max_{i=1,\dots,n} w_i |x_i|$ is a norm on $\mathbb{R}^n$.

Subordinate Matrix Norm

A subordinate matrix norm is the largest factor by which matrix can stretch a vector measured by a vetor norm.

Problem 2

To prove that $\|A\|_1 := \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{j=1,\dots,n} \sum_{i=1}^n |a_{ij}|$, the definition of an induced matrix norm needs to be re-visited.

$$\|A\| = \sup_{\|u\|=1} \{Au\}$$

By definition of a supremum, the induced norm of A needs to be greater than or equal to all elements in the resulting dot product for any unit vector u , ie an upper bound. In order to find this, let u be a vector of all zeros except for some index j being equal to 1 and thus satisfies $\|u\|_1 = 1$. By computing the dot product of Au , the j^{th} column is extracted. The norm of the resulting vector is $\sum_{i=1}^n |Au_i|$. In order to find an upper bound, multiply the matrix by a set of vectors u where each vector in the set is zero except some index j equal to 1. No vectors in the set have the same non zero j . For each vector, compute the dot product and the 1-norm. Once all norms have been calculated, take the maximum norm as the upper bound of the induced matrix 1-norm. Therefore, the upper bound, or supremum, of $\|A\|_1$ is $\max_{j=1,\dots,n} \sum_{i=1}^n |a_{ij}|$.

Problem 3

Part 1

To show that A is non-singular by $\|Ax\| \geq \theta \|x\|$, $\forall x$, first set x to zero. Since $\|\cdot\|$ is a norm on $\mathbb{R}^n$, x needs to be zero in order for the norm to be zero. This means that the $\ker(A) = \{0\}$. Because the the kernel of A is only the zero vector, the matrix is invertible, a.k.a non-singular.

Since A is invertible, to show that $\|A^{-1}\| \leq \theta^{-1}$, say there is some vectopr y , where $Ax = y$ and $A^{-1}y = x$. Since $\|Ax\| \geq \theta \|x\|$, then $\|y\| \geq \theta \|A^{-1}y\|$. Rearranging both sides, the expression is as follows, $\frac{1}{\theta} \geq \frac{\|A^{-1}y\|}{\|y\|}$, which can be rewritten as $\frac{1}{\theta} \geq \sup\{\|A^{-1}y\|\}$, or more simply as $\theta^{-1} \geq \|A^{-1}\|$.

Part 2

To show that A is invertible based on $\|I - P(A)\| \leq 1$, the Neuman series can be used. This is because the norm of $A - I$ is less than 1. The series is as follows:

$$(I - A)^{-1} = \sum_{i=0}^{\infty} A^i$$

Substituting A with $I - P(A)$ gives $P(A)^{-1} = \sum_{i=0}^{\infty} (I - P(A))^i$. Since P is a polynomial with no constant terms, A can be factored out of the matrix, resulting in $(AG(A))^{-1} = \sum_{i=0}^{\infty} (I - P(A))^i$, where $G(A)$ is the resulting polynomial with one constant term after A is factored out. Rewritting the expression, $A^{-1} = G(A) \sum_{i=0}^{\infty} (I - P(A))^i$.

Problem 4

part 1

If A is positive symmetric definite, then its diagonals must be > 0 . Say there is a unit vector e_k , where all entries are zero except for the entry at index k , which is set to 1. using the quadratic form, $\langle e_k A, e_k^T \rangle$, The only influencing entry in the matrix that determines whether the scalar result is positive is the k^{th} diagonal. If the quadratic form is computed from $k = 1$ to n , the products should all be > 0 .

part 2

Say there is some vector $x \in \mathbb{R}^n$ and on index i , $x_i = \alpha$ and on index j , $x_j = 1$ with zeroes everywhere else. By using the quadratic form $\langle Ax, x^T \rangle$, the following vector and equation are generated.

$$Ax = ((\alpha a_{1i} + a_{1j}), \dots, (\alpha a_{ii} + a_{ij}), \dots, (\alpha a_{ji} + a_{jj}), \dots, (\alpha a_{ni} + a_{nj}))$$

$$\langle Ax, x^T \rangle = \alpha^2 a_{ii} + \alpha a_{ij} + \alpha a_{ji} + a_{jj} = \alpha^2 a_{ii} + 2\alpha a_{ij} + a_{jj}$$

When factoring the equation above, the roots of the polynomial are $a_{ii}(\alpha + \frac{a_{ij}}{a_{ii}})^2 + (a_{jj} + \frac{a_{ij}^2}{a_{jj}}) = (a_{jj} - \frac{a_{ij}^2}{a_{ii}}) > 0$. Rearranging the equation gives $a_{ii}a_{jj} > a_{ij}^2$.